

CLASS-PRIOR CONSTRAINTS ON CONFIDENT-TAIL TEST-TIME ADAPTATION FOR AUDIO DEEPPAKE DETECTION

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ABSTRACT

Cross-corpus failure in audio deepfake detection is often a failure of the decision threshold rather than of score ordering: a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but 39.08% accuracy at threshold 0.5. Confident-tail self-training exploits the surviving ordering by pseudo-labeling the lowest- and highest-scoring target clips, and we ask when that rule helps or harms. A bucket assigned class c and occupying fraction b_c of a pool with prior π_c cannot be purer than $\min(1, \pi_c/b_c)$; equivalently it must carry at least $N(b_c - \pi_c)^+$ wrong labels. Across 31 adaptation runs spanning four released checkpoints, two corpora and spoof prevalence 0.01 to 0.97, that excess mass, and not purity, orders the outcomes: below an empty gap 18 of 20 runs improve balanced accuracy by 5.7 to 32.7 points, while above it none of 11 improves by more than 4.3 and eight are harmful. Accuracy at a fixed threshold exaggerates the effect. On ASVspoof2021-DF, where always predicting spoof already scores 97.22%, a nominal $q = 0.3$ budget moves accuracy from 98.87% to 62.96% but balanced accuracy only from 88.28% to 80.89% and EER from 4.51% to 5.84%. A black-box label-shift budget repairs the high-prevalence case, yet estimates 35.9–41.8% spoof on pools that are 1.1–9.7% spoof, where only a small prior-free budget helps.

Index Terms— audio deepfake detection, test-time adaptation, label shift, pseudo-labels

1. INTRODUCTION

Self-supervised speech representations paired with anti-spoofing back-ends provide strong audio deepfake detectors [1, 2], but transfer across recording conditions and synthesis systems remains difficult [3]. A target-domain failure can involve both score ordering and the decision threshold. For example, a released detector in our In-the-Wild evaluation has ROC-AUC 0.9171 but only 39.08% accuracy at threshold 0.5, which motivates examining adaptation and rethresholding separately without assuming that ranking always transfers.

Test-time adaptation (TTA) uses unlabeled target observations to update a source model. Entropy minimization [4] and pseudo-label self-training can reinforce incorrect predictions. We examine a specific self-training rule: assign the bottom and top score quantiles to bona fide and spoof classes. This symmetric-quantile rule is not one proposed elsewhere in the literature; it is our own prior design, an instance of confidence-based pseudo-label selection [5, 16] with a fixed budget on each tail. What follows is therefore an analysis of a rule we had adopted ourselves. With a fixed symmetric budget the minority bucket can exceed the entire minority population, so no ranking can fill it correctly, which matters precisely when benchmark and deployment pools differ in class proportions.

Class imbalance in adaptation is already established: Label Shift Adapter [6] estimates target proportions to adapt model parameters, and RLSbench [7] studies adaptation under joint label and conditional shift; black-box shift estimation (BBSE) [8] estimates target priors from a source confusion matrix, and T²A [9] addresses deepfake TTA through uncertainty-aware negative learning. Our contribution is narrower: a counting constraint for confident-tail sampling, the quantity in which it binds, and an audio checkpoint study of its consequences. We claim no new label-shift estimator and no superiority to specialized label-shift adaptation.

2. TAIL SELECTION AND ADAPTATION

Let $y = 1$ denote spoof and $s_\theta(x) \in [0, 1]$ the detector’s spoof score. The unlabeled adaptation pool has N clips and unknown class fractions π_c . Source labels are available for BBSE; target labels are used only for evaluation, controlled resampling, and explicitly marked oracle analyses.

2.1. A necessary condition for pure pseudo-labels

A bucket C_c assigned pseudo-label c , with realized fraction $b_c = |C_c|/N > 0$, contains at most $\pi_c N$ correct examples. Thus

$$\text{purity}(C_c) \leq \min\{1, \pi_c/b_c\}. \quad (1)$$

This is a finite-pool counting bound and makes no assumption about the model. An update, however, does not consume a purity: it consumes wrongly labeled examples. The same counting fact states that C_c contains at least

$$W_c = N(b_c - \pi_c)^+ \quad (2)$$

of them. Purity is a rate and W_c a mass, and the two order configurations differently: a small bucket can have low purity while contributing few wrong labels. Section 4 finds that $\sum_c W_c$ orders the observed outcomes and that purity does not.

For exact symmetric tails, $b_0 = b_1 = q$, both statements reduce to $\sum_c W_c = N(q - \min\{\pi, 1 - \pi\})^+$, so perfect purity is possible only if

$$q \leq \min\{\pi, 1 - \pi\}, \quad \pi = \pi_1. \quad (3)$$

At 97.22% spoof prevalence a bucket labeling 30% of the pool bona fide has purity at most 9.27%, and no amount of AUC removes that. Feasibility is not sufficiency: score overlap contaminates feasible tails, and learning tolerates impure labels when there are few of them. That is why we test (2) rather than (3).

A prior allocation uses nominal fractions $q_c = 2q\hat{\pi}_c$, keeping the total selection fraction $2q$. With exact buckets and a correct prior, $q \leq 1/2$ ensures $q_c \leq \pi_c$; with estimated priors it instead requires $\hat{\pi}_c/\pi_c \leq 1/(2q)$, a limit on overestimation rather than a safety theorem. A smaller symmetric q needs no point estimate, but a guaranteed feasible nonzero value still needs a lower bound on the minority prevalence.

Quantile ties. Our cached experiments use inclusive score thresholds, $s \leq Q(q_0)$ and $s \geq Q(1 - q_1)$, with spoof assignments taking precedence if masks overlap. Consequently, realized sizes can differ from nominal budgets. Equations (1) and (2) apply to the realized b_c ; the conditions above it describe exact-size tails. We audit realized selections on cached source scores and do not interpret nominal budgets as exact sample counts. Deterministic selection of fixed counts would be a different implementation requiring new adaptation runs.

2.2. Prior estimation and objective

For $M_{ij} = P_S(\hat{y} = j \mid y = i)$ and target prediction histogram h , BBSE solves $M^\top \hat{p} = h$. We estimate M using up to 4,000 class-balanced clips from the checkpoint’s labeled training corpus; h uses source predictions on the adaptation pool. We clip the solution components to $[10^{-4}, 1 - 10^{-4}]$ and renormalize. The implementation returns a uniform prior if the matrix is singular. Because M uses training rather than held-out source data, optimism in its estimated error rates is a limitation.

At each of four passes, the current model scores the adaptation pool and refreshes pseudo-labels. For a mini-batch B

and its selected subset B_C ,

$$\begin{aligned} \mathcal{L}_B = & \frac{1}{|B_C|} \sum_{x \in B_C} \text{CE}(p_\theta(x), \tilde{y}_x) \\ & + \frac{\lambda}{2|B|} \sum_{x \in B} \|p_\theta(a(x)) - \text{sg}[p_\theta(x)]\|_2^2, \end{aligned} \quad (4)$$

where $p_\theta = (1 - s_\theta, s_\theta)$, the CE term is omitted when B_C is empty, and $\lambda = 0.3$. The consistency term uses all batch examples and both probability components. The perturbation is per-clip gain sampled uniformly from $[0.7, 1.3]$ plus Gaussian noise with standard deviation 0.005. Only the last four transformer blocks’ LayerNorm affine parameters and the classifier head are updated; BatchNorm statistics stay frozen. Adam uses learning rate 10^{-4} and batch size 32. The released DeepFense models have 16,706 trainable parameters under this rule. A fixed-confidence control instead selects scores at most 0.05 or at least 0.95, with the same objective.

3. EVALUATION PROTOCOL

Checkpoints and pools. We use DeepFense XLS-R 300M + AASIST checkpoints trained on ASVspoof2019-LA with training seeds 2, 42, and 240 [10, 11], denoted DF-2, DF-42, and DF-240. A WaveFake-trained SSL-AASIST checkpoint, WF [12], is evaluated on In-the-Wild. Audio is 16 kHz with the checkpoint-specific input length: 64,000 samples for DeepFense and 64,600 for WF.

The official ASVspoof2021-DF metadata [13] has 533,928 eval trials among 611,829 across three phases. Our available *eval-only* subset is 400,435 of them, 389,299 spoof and 11,136 bona fide, or 97.219% spoof, with 75.00% class coverage of spoof and 74.89% of bona fide. The missing shard rules out a full-set comparison; all method comparisons use the same trials.

The main run adapts on 20,000 randomly selected clips with adaptation seed 0. We report both the available pool, which includes adaptation clips, and its 380,435-clip disjoint remainder. Additional adaptation seeds 1 and 2 use a fixed 50,000-trial subset, selected with seed 12345, with 20,000 clips for adaptation per run. These are sensitivity checks on overlapping data, not nine independent datasets. In-the-Wild uses all 31,779 clips (11,816 spoof; 37.182%), with 20,000 adaptation clips and an 11,779-clip disjoint remainder.

Prevalence intervention. On In-the-Wild with WF, we resample the pool to target spoof proportions (sampling seed 4242): above the native 0.372 we retain all spoof clips and thin bona fide clips, and below it we retain all bona fide clips and thin spoof clips. Every resampled pool adapts on 8,000 clips, so the number of optimization steps is identical at every prevalence and a disjoint remainder is always available for evaluation. The evaluation pool size still varies with prevalence, and the experiment does not separate that from prevalence itself.

Table 1. Recomputed results on the 400,435 available DF eval trials, checkpoint DF-2, adaptation seed 0. Acc. and balanced accuracy (BA) use threshold 0.5 except for the two frozen-source threshold controls. G_τ is an oracle diagnostic in percentage points; smaller is better. Baselines are the specified variants in Sec. 3.

Method	EER (%)	AUC	Acc. (%)	BA (%)	G_τ
Source	4.51	0.9923	98.87	88.28	0.20
Entropy (Tent-style)	4.28	0.9765	98.52	73.51	0.57
Fixed confidence 0.95	4.30	0.9918	99.02	85.03	0.00
Symmetric $q = 0.02$	4.19	0.9930	98.96	83.88	0.01
Symmetric $q = 0.3$	5.84	0.9846	62.96	80.89	35.38
BBSE tail budget	4.06	0.9931	98.95	82.34	0.02
Source + median threshold	4.51	0.9923	59.06	78.94	40.01
Source + BBSE quantile threshold	4.51	0.9923	98.88	88.19	0.19

Metrics and controls. We recompute metrics from saved float32 probabilities. ROC-AUC assigns half credit to ties; EER linearly interpolates the crossing of FPR and FNR on the ROC, grouping identical scores. Interpolated EER need not correspond to an attainable deterministic threshold. We also report accuracy, balanced accuracy, and

$$G_\tau = \max_t \text{Acc}(s, t) - \text{Acc}(s, \tau). \quad (5)$$

The maximum ranges over all distinct thresholds, including constant predictions. It uses evaluation labels and is a diagnostic of threshold headroom, not a deployable method or a calibration error.

Two label-free controls keep source scores fixed and pick a threshold from adaptation scores: their median, or their $(1 - \hat{\pi})$ -quantile using BBSE. Both preserve AUC and EER; neither is a claim of Bayes optimality.

We additionally report fixed-configuration entropy adaptation (Tent-style) under the same update settings [4].¹

4. RESULTS

4.1. Preventing large-budget damage on DF

Table 1 shows that symmetric $q = 0.3$ lowers DF-2 accuracy from 98.87% to 62.96% and increases EER from 4.51% to 5.84%. Its oracle threshold gap is 35.38 points: much of the fixed-threshold damage is recoverable by re-thresholding, although AUC also declines. The BBSE budget reaches 98.95% accuracy and 4.06% EER. Fixed confidence reaches higher accuracy (99.02%), and the smaller $q = 0.02$ reaches 98.96% accuracy and 4.19% EER. Prior allocation is therefore one effective remedy, not a uniformly best choice. Balanced accuracy is lower after BBSE adaptation than for the source

¹Three further published TTA methods were run the same way, as controlled variants rather than tuned reproductions: an ETA reliability-only ablation without redundancy or Fisher terms (EER 12.31%, and an all-spoof decision at threshold 0.5 on DF-42 and DF-240), SAR with recovery threshold scaled to $(0.2/\ln 1000)\ln 2$ (4.48%), and IM-PL, a SHOT-inspired information-maximization control with probability-space centroid pseudo-labels that freezes the head and updates 16,384 parameters (6.35%) [14–16]. None is a tuned reproduction and none establishes anything about those methods’ published performance; their audited rows are in the released tables.

Table 2. DF checkpoint comparison on the same available eval pool. Entries are EER%/accuracy%; symmetric and BBSE budgets use nominal $q = 0.3$.

Checkpoint	Source	Sym.	BBSE
DF-2	4.51/98.87	5.84/62.96	4.06/98.95
DF-42	4.56/98.78	4.10/61.80	3.30/98.98
DF-240	3.82/99.11	4.66/65.24	3.80/98.74

(82.34% versus 88.28%); the slight overall accuracy gain should not be interpreted as an improvement for both classes.

The accuracy collapse and its reversal occur on all three checkpoints (Table 2). Across the nine checkpoint/run combinations, BBSE has lower EER than the symmetric budget in 9/9 on the available pool and 9/9 under disjoint evaluation, and lower than the source in 8/9 and 7/9 respectively. A paired stratified bootstrap over trials separates the two comparisons: 2,000 replicates resampled within class, so each replicate holds the pool’s prior fixed. All nine symmetric deltas have 95% intervals excluding zero, but only 4/9 of the source deltas do on the available pool and 5/9 under disjoint evaluation. The DF-240 difference is -0.02 points ($[-0.14, +0.12]$) on the available pool and $+0.003$ ($[-0.12, +0.15]$) under disjoint evaluation; neither is evidence of a gain. We give the counts as descriptive, without treating reused checkpoints and trials as independent evidence. Preventing severe damage is thus far better supported than improving an already strong source.

The frozen-source BBSE threshold gives 98.88% accuracy for DF-2, close to its 98.87% source result, while the median gives 59.06%. Median thresholding implicitly targets roughly balanced predictions and is poorly suited to this pool. The budget change can improve EER, which threshold changes cannot do, but it is not required to preserve high accuracy here. Removing consistency from the BBSE objective gives 3.98% EER versus 4.06% with it on DF-2; this single ablation provides no evidence that consistency is needed for the gain.

Score resolution. Tent-style adaptation places 98.07% of its scores within 10^{-6} of 1.0 but only 93.02% exactly at

Table 3. WF on In-the-Wild resampled to spoof prevalence π , EER%/balanced accuracy%. Balanced accuracy has the same 50% trivial baseline in every row, whereas the always-spoof accuracy baseline moves from 90.00% to 99.00% across them. Adaptation uses 8,000 clips at every π , so the optimization budget is matched; a dash marks a budget not run at that prevalence.

π	N	Source	$q = .02$	$q = .10$	$q = .30$	BBSE
0.372	31,779	5.06/82.7	—	—	4.55/95.4	4.54/95.5
0.010	20,165	5.10/82.7	4.95/92.4	5.10/92.1	5.27/75.7	5.17/82.2
0.020	20,370	5.16/82.7	5.00/92.9	5.12/91.8	5.23/79.0	5.16/82.7
0.050	21,014	5.14/82.6	5.04/93.7	5.11/94.6	5.09/78.8	5.10/82.6
0.100	22,181	5.12/82.7	5.04/94.0	4.96/94.5	5.05/82.3	5.02/85.2
0.900	13,129	5.26/82.7	5.04/93.7	5.26/95.0	6.02/84.4	5.03/93.7
0.950	12,438	4.66/83.6	4.58/92.5	4.94/94.6	5.66/83.0	4.52/91.6
0.970	12,181	3.74/83.5	3.91/93.3	4.66/93.7	5.21/81.8	3.84/92.3

1.0; conflating the two would make its AUC appear impossible. With exact ties grouped it reaches EER 4.28% and AUC 0.9765 against source 4.51% and 0.9923, so an improvement near one ROC operating point can coexist with worse integrated ranking. The scoring convention itself matters: for the ETA ablation, ROC interpolation gives EER 12.31% where a nearest-ROC-point rule would give 7.44%. This supports reporting several metrics, not dismissing EER as arbitrary tie-breaking.

4.2. Corpus shift and prior estimation

At the native In-the-Wild prior the symmetric budget has no counting obstruction, and it improves accuracy on three of the four checkpoints; on DF-2 it does not, feasibility again being insufficient. For DF-42 it raises AUC from 0.9171 to 0.9473 and accuracy from 39.08% to 82.39%, though a frozen-source median threshold already reaches 78.10%: the comparison depends heavily on the operating point. BBSE estimates 97.9% spoof on this 37.2%-spoof corpus and lands at 14.54% EER and 47.99% accuracy, against 11.59%/91.28% for an oracle budget that is not deployable and not always best, as on WF the oracle gives 4.67% EER against 4.55% for the symmetric rule. BBSE is identifiable under label shift with an invertible confusion matrix, even for uncalibrated predictors [8]; under unrestricted conditional shift the prediction rate $P_{\mathcal{T}}(\hat{y} = 1) = \pi \text{TPR}_{\mathcal{T}} + (1 - \pi) \text{FPR}_{\mathcal{T}}$ alone cannot determine all three unknowns, which is why a source confusion matrix may fail to transfer. That is not an impossibility result for output-based estimators with further assumptions.

4.3. Intervening on prevalence

The large-budget degradation also occurs on resampled In-the-Wild (Table 3). At 97% spoof, nominal $q = 0.3$ permits at most 10% purity in an exact-size bona fide tail; its accuracy is 65.42% against 98.59% for $q = 0.02$ and 98.94% for

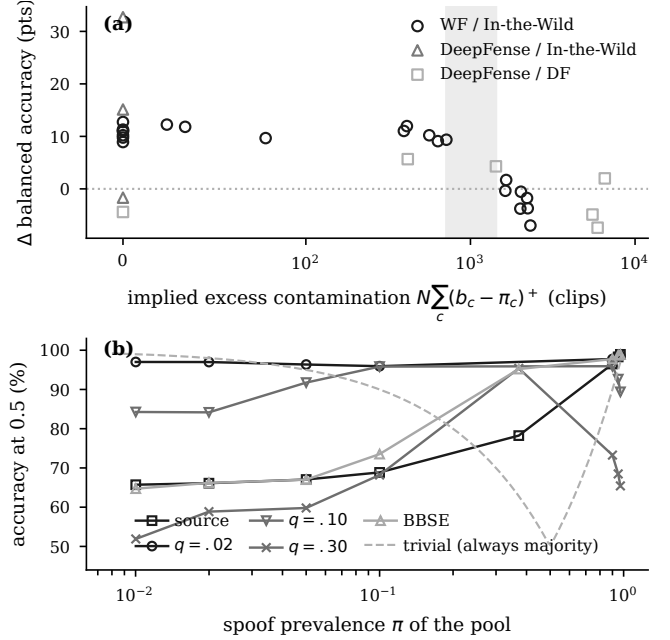


Fig. 1. (a) Balanced-accuracy change from symmetric-tail adaptation against the excess contamination (2) implies, over 31 runs spanning four checkpoints, two corpora and prevalences 0.01–0.97. No run lands in the shaded gap. (b) Accuracy at threshold 0.5 on resampled In-the-Wild, against the always-majority baseline (dashed), which is 99.00% at $\pi = 0.01$ and 97.00% at $\pi = 0.97$.

the unchanged source. Which budget looks better depends on the metric, and the earlier claim that a small budget prevents the collapse without improving the source holds only for accuracy at threshold 0.5. In balanced accuracy the same cells read 81.8% for $q = 0.3$, 93.3% for $q = 0.02$ and 83.5% for the source, so the small budget is a substantial improvement; in EER it is 5.21%, 3.91% and 3.74%, so it is a slight regression. Accuracy at a fixed threshold on a 97%-spoof pool is the metric that moves most and means least: the always-spoof rule alone scores 97% there.

The deployment direction. Real pools are mostly bona fide, so we extend the sweep to $\pi \in \{0.01, 0.02, 0.05, 0.10\}$, where the constrained tail is the spoof one. The mirror of the high-prevalence collapse appears: at $\pi = 0.01$ nominal $q = 0.3$ moves accuracy from 65.72% to 51.90% and balanced accuracy from 82.69% to 75.71%, and the damage is carried entirely by false alarms (FPR 48.58% against 34.62%; FNR 0.00% in both). The small budget, however, does not break where (3) says it must: at $\pi = 0.01$ its spoof tail is bounded at 52.7% purity and measured at 33.3%, yet it reaches 97.00% accuracy and 92.36% balanced accuracy. Ranking barely moves anywhere in the sweep: every arm at every prevalence has EER between 4.95% and 5.27% while accuracy ranges from 51.90% to 97.00%.

What orders the outcomes. Fig. 1(a) pairs each symmet-

ric cell’s implied excess $\sum_c W_c$ with the balanced-accuracy change adaptation produced. No cell lands between 715 and 1,427 excess clips. Below that gap 18 of 20 cells improve balanced accuracy by 5.7 to 32.7 points; above it none of 11 improves by more than 4.3 and eight are harmful. Worst-tail purity does not order the same cells: 0.107 gives +9.4 points and 0.980 gives −4.4. The barrier is one-sided. The two exceptions below the gap are exactly what (1) cannot rule out, since feasible selection is not sufficient, whereas no cell above the gap recovered. The separation is unchanged under disjoint evaluation. It rests on one adaptation recipe and reused audio, and the threshold’s location inside the empty gap is not identified.

Prior estimation in that direction. BBSE, which repaired the high-prevalence collapse, does not transfer here: it estimates 35.9% to 41.8% spoof on pools that are 1.1% to 9.7% spoof, and changes accuracy by −1.0 to +4.7 points against the source, where the prior-free $q = 0.02$ budget gains 27.1 to 31.3. A small budget is the only rule in this study that helps at both ends of the prevalence range.

5. DISCUSSION AND LIMITATIONS

The counting obstruction gives a useful diagnostic for pseudo-label allocation, while its consequences depend on score quality, realized selection sizes, and the learning objective. Recomputed metrics support severe fixed-threshold damage from large symmetric tails and its mitigation by prior allocation or smaller tails. They also show important tradeoffs: some remedies reduce minority-class performance, prior estimates can fail on shifted corpora, and re-thresholding alone explains a substantial part of some accuracy gains.

The experiments use four released checkpoints from one broad architecture family, two principal corpora, an incomplete DF eval partition, and fixed adaptation hyperparameters. They do not establish general superiority over specialized label-shift methods. Our prevalence intervention changes sample count, and the seed checks share data. Future validation should use independent corpora, fixed-size prevalence interventions, held-out source calibration, exact-count tail selection, and development-domain tuning of baseline methods. Neither a small nominal tail nor source-score agreement alone certifies safe adaptation without assumptions about the target distribution.

Reproducibility. The script `audit_icassp.py` verifies 89 saved score arrays against historical AUC and accuracy, recomputes the reported metrics, and writes the tables and input hashes. The project repository is https://github.com/RohamIzadidoost/Kaggle_deepFakeAudio. The accompanying audit bundle includes the revised scripts and outputs. No new neural-network training is required to reproduce these tables from the saved scores and original manifests.

Ethics and assistance. This study uses existing public datasets and checkpoints; no new human-subject data

were collected. OpenAI Codex assisted with the revised manuscript text, metric-audit code, and result tables; the adaptation experiments predate this revision.

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